8 SARMs PROFILED	MECHANISM GUIDE	STACK PROTOCOLS	MINI-PCT

SARMs

COMPLETE SCIENTIFIC HANDBOOK

RAD-140 · LGD-4033 · Ostarine · Cardarine · YK-11 · MK-677 · S23 · Full Stacking Guide

Mechanism of action · Tissue selectivity · Dosing · Mini-PCT · Side effect management

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■ 1. WHAT ARE SARMs? — THE SCIENCE

Selective Androgen Receptor Modulators (SARMs) are a class of drugs designed to activate androgen receptors in a **tissue-selective** manner — producing anabolic effects in muscle and bone while minimizing androgenic effects in other tissues (prostate, skin, liver, etc.).

> ■ *SARMs were originally developed in the 1990s by pharmaceutical companies as treatments for muscle wasting diseases, osteoporosis, and male hypogonadism. None have received FDA approval for these uses, but extensive research exists confirming their anabolic activity.*

How Tissue Selectivity Works:

- SARMs are non-steroidal compounds that bind the androgen receptor (AR) with high affinity
- Unlike testosterone, SARMs produce different **conformational changes** in the AR depending on the tissue
- In muscle: SARM-AR complex recruits coactivator proteins that drive anabolic gene expression
- In prostate/scalp: SARM-AR complex recruits different coactivators with weaker or no androgenic activity
- Result: muscle-building effects similar to testosterone with fraction of androgenic side effects

Compound	Anabolic Ratio	Androgenic Ratio	Liver Toxicity	Suppression Level
Testosterone (baseline)	100	100	Low (injectable)	Complete (expected)
RAD-140	~890	~3	None (non-17aa)	Moderate–High
LGD-4033	~500	~4	None	Moderate
Ostarine	~30	~1	None	Mild–Moderate
YK-11	Extremely high*	Moderate	Unknown	High
S23	~830	~98	None	Very High
MK-677	N/A (not true SARM)	None	None	None (raises GH)

■ 2. RAD-140 (TESTOLONE)

RAD-140 is the most anabolic SARM ever developed — with an anabolic:androgenic ratio of approximately 90:1 compared to testosterone's 1:1. Originally developed by Radius Health for muscle-wasting diseases, it produces **significant lean mass gains comparable to low-dose testosterone**.

Parameter	Details
2S)-2-hydroxy-2-methyl-1-[4-(trifluoromethyl)phenyl]propyl]amino]-3-methylbenzonitrile	
60 hours (dose once daily)	
90:1 (vs testosterone 1:1)	
8–15 mg/day	
8–12 weeks	
Moderate–High (mini-PCT required)	
Lean muscle mass, strength, body recomposition	
MK-677, Cardarine, LGD-4033 (bulking stack)	

- Expected results: 8–12 lbs lean mass over 12 weeks (first cycle); superior to most oral steroids
- Does not aromatize — no estrogen-related side effects, no AI required
- May cause hair shedding in DHT-sensitive individuals (binds scalp AR despite selectivity claims)
- Neurological benefits: RAD-140 is neuroprotective in animal studies; enhanced mood and focus commonly reported
- Blood work: Check testosterone and LH/FSH at baseline and Week 8; expect 30–60% suppression

■ 3. LGD-4033 (LIGANDROL)

LGD-4033 (Ligandrol) was developed by Ligand Pharmaceuticals and studied clinically in Phase I trials. It is the **most studied SARM in human clinical trials**, with a published Phase I safety study at doses of 0.1–1 mg/day showing significant lean mass gains even at microdose levels.

■ *Basaria et al. (2013, The Lancet) Phase I trial: LGD-4033 at 1 mg/day for 21 days produced a statistically significant 1.21 kg lean mass gain with dose-dependent suppression of LH, FSH, and total testosterone. At 1 mg/day, no serious adverse events were reported.*

- **Dose:** 5–10 mg/day (5mg is sufficient for lean bulking; 10mg for aggressive bulking)
- **Half-life:** 24–36 hours — once daily dosing
- **Best use:** 8–12 week bulking or recomposition cycles
- **Expected results:** 10–15 lbs lean mass over 12 weeks at 10mg/day with proper diet
- **Water retention:** Moderate (1–3 lbs) — more than RAD-140 due to mild estrogenic activity
- **Suppression:** Moderate — mini-PCT recommended; recovery typically complete in 4–6 weeks

■ 4. OSTARINE (MK-2866)

Ostarine (Enobosarm) is the mildest and most studied SARM — the safest entry point for SARM use. It was developed by GTx Inc. and studied extensively in Phase II trials for cancer cachexia and osteoporosis. Its mild profile makes it ideal for beginners, recomposition, and joint healing.

- **Dose:** 15–25 mg/day for mass/recomp; 10–15 mg for joint healing
- **Half-life:** 23–26 hours — once daily dosing
- **Suppression:** Mild — many users need no PCT after 8-week cycles at 15–20mg
- **Joint healing:** Demonstrated improvement in collagen synthesis and joint integrity markers
- **Female-friendly:** 5–10 mg/day is one of the safest anabolic compounds for women
- **Stacking:** Excellent base for all stacks; pairs well with Cardarine for recomp

■ 5–8. CARDARINE · YK-11 · MK-677 · S23

■ CARDARINE (GW-501516)

PPAR-δ (Peroxisome Proliferator-Activated Receptor Delta) agonist — technically NOT a SARM but classified with them. Dramatically increases fat oxidation and endurance by switching cells from glucose to fat metabolism.

■■ **IMPORTANT:** CRITICAL WARNING: GW-501516 was abandoned by GlaxoSmithKline in 2007 when animal studies showed it caused cancerous tumors at extremely high doses (>10× human dose for 2 years). Use at your own risk; keep doses low and cycles short.

Protocol: 10 mg/day (4–8 weeks maximum). Benefits: endurance increases 30–60%; LISS cardio capacity dramatically improved; fat loss accelerated. Do NOT exceed 8-week cycles.

■ YK-11

Partially activates androgen receptor (SARM activity) AND inhibits Myostatin — the body's natural muscle growth limiter. Theoretically allows muscle mass beyond genetic ceiling. YK-11 is the most experimental compound in this guide.

■■ **IMPORTANT:** Extremely limited human data exists. Mechanism is promising but safety profile is largely unknown. Animal data shows liver enzyme elevation at high doses. NOT recommended for beginners.

Protocol: 5–10 mg/day, 6 weeks maximum. High suppression — full PCT required. Liver support (TUDCA, NAC) mandatory.

■ MK-677 (IBUTAMOREN)

Ghrelin receptor agonist and GH secretagogue — increases GH and IGF-1 WITHOUT suppressing the HPT axis. Not a true SARM. Dramatically improves sleep quality, recovery, skin, hair, and produces moderate anabolic effects over months of use.

■■ **IMPORTANT:** 25 mg nightly before bed (capitalizes on nocturnal GH pulse). Benefits: GH increases 30–100%; IGF-1 rises 50–100%; dramatically improved deep sleep; noticeable joint and connective tissue improvement over 3+ months. Side effects: water retention, increased appetite, potential insulin resistance (monitor fasting glucose).

Protocol: Cycle length: Can be run long-term (6–12 months). No PCT required. Stack with any compound safely.

■ S23

The most potent and most suppressive SARM available. Almost identical androgenic activity to anabolic steroids. Produces extreme hardness, vascularity, and dry lean mass gains. Studied as a male contraceptive — causes complete suppression of sperm production at certain doses.

■■ **IMPORTANT:** Full PCT required (same as steroid cycle). NOT a beginner compound.

Protocol: 10–30 mg/day, 6–8 weeks. Expect suppression at 100% of testosterone production. Use lowest effective dose. Excellent for cutting/recomp in experienced users.

■ 9. STACK PROTOCOLS

Goal	Stack	Doses	Duration	PCT
Lean Bulk	RAD-140 + MK-677	RAD-140 10mg/day + MK-677 25mg nightly	12 weeks	Mini-PCT 4 weeks
Aggressive Bulk	LGD-4033 + MK-677 + RAD-140	LGD 10mg + RAD 10mg + MK-677 25mg	10 weeks	Full PCT 6 weeks
Cutting/Recomp	Ostarine + Cardarine	Ostarine 20mg/day + Cardarine 10mg/day	8 weeks	Optional mini-PCT
Advanced Cutting	RAD-140 + Cardarine + S23	RAD 10mg + GW 10mg + S23 15mg	8 weeks	Full PCT required
Women's Recomp	Ostarine only	10 mg/day	8 weeks	None needed at this dose
Joint Healing	Ostarine + BPC-157	Ostarine 15mg/day + BPC-157 250mcg 2×/day	8 weeks	None / mini-PCT

■ 10–12. MINI-PCT · BLOOD WORK · SARMs vs AAS

Mini-PCT for SARMs (most cycles):

- Clomid 25–50 mg/day for 4 weeks (lower dose than AAS PCT) + Nolvadex 20 mg/day for 4 weeks
- Start mini-PCT 24–48 hours after last dose (SARMs have short half-lives vs AAS long esters)
- For S23, RAD-140 at high dose, or YK-11: use full AAS-level PCT protocol (Clomid 50mg + Nolvadex 20mg for 6 weeks)

• MK-677: no PCT required — does not suppress HPT axis

Minimum Blood Work:

Timing	Tests
Pre-SARMs	Total T, Free T, LH, FSH, E2, CBC, CMP, Lipids
Week 6 on-cycle	Total T, LH, FSH (check suppression level)
Post-PCT (4 wks)	Total T, LH, FSH, E2 — confirm recovery

SARMs vs AAS — Key Differences:

Factor	SARMs	Anabolic Steroids
Mechanism	Non-steroidal AR modulation	Steroidal AR activation
Liver toxicity	Generally none (non-17-alpha alkylated)	Orals: significant; injectables: minimal
Estrogen conversion	None (except LGD: mild)	Yes — varies by compound
Hair loss risk	Lower (selective)	Higher (DHT conversion)
Cardiovascular impact	Moderate (lipid changes)	Significant (HDL suppression)
PCT complexity	Mini-PCT usually sufficient	Full PCT typically required
Legal status	Unscheduled research chemical (most countries)	Schedule III controlled substance (USA)

■ *SARMs are not approved for human use by any regulatory agency. For research purposes only. Consult a physician.*